

Nathan Tebbs

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SKILLS

Languages: C/C++, Java, Python, Go, Rust, Odin, JavaScript, TypeScript, Lua, SQL

Tools: Git/GitHub, Docker, Linux, make, cmake, systemd, GDB, Valgrind, SSH, VMware/KVM, Neovim & Emacs, Vercel

Libraries: SDL3, Raylib, pandas, NumPy, scikit-learn, Streamlit, React, Auth0

EDUCATION

University of Arizona

Bachelor of Science in Computer Science

• Relevant Coursework: Compilers, Database Design, Algorithms, Computer Networking, Cloud Computing, Systems Programming

South Puget Sound Community College

Associate of Arts

Tucson, Arizona

August 2023 — May 2026

Olympia, Washington

September 2021 — May 2023

WORK EXPERIENCE

Software Engineer (Full-Stack)

Sound Psychiatric Services

January 2021 — Present

Olympia, Washington

- Build and maintain the practice's public website, a landing page for around 100-120 patients that routes existing patients to their HIPAA-compliant payment and telehealth portals and informs prospective patients before an in-person or Zoom visit.
- Migrated hosting off a paid Squarespace business plan to a static setup, saving the practice over \$5,000 a year, and kept the existing email and domain in place so the transition caused no disruption. Built in plain HTML, CSS, and JavaScript to stay simple and low-maintenance.

Undergraduate Teaching Assistant (CSc 346 Cloud Computing)

University of Arizona

January 2025 — May 2025

Tucson, Arizona

- Assisted students with Linux-based cloud VMs, web servers, networking, and deployment issues.
- Troubleshoot configuration errors and documented common failure modes and resolutions.
- Worked with other teaching staff to meet course timelines on grading and office hours.

PROJECTS

Author, Mach Game Engine (github.com/vetr0s/mach)

June 2026 — Present

- Building a 2D game engine and the game it runs, Factory, together in pure C and SDL3. Factory is an isometric belt-builder: drop ore, route it through upgraders that raise its value, and bank it to grow the grid.
- Wrote the core subsystems from scratch, including an arena allocator, vector and matrix math, a 2D isometric renderer on `SDL_Renderer`, a bitmap font atlas, and a fat-struct entity system, all built as a unity build with one `clang` call and no separate build system.
- Built live hot reloading for development: the game compiles to a shared library the running host reloads on save, and because all state lives in host-owned memory, new code swaps in with no restart and no lost state. Release builds go through each platform's native toolchain with SDL3 bundled alongside, so one build runs on macOS, Linux, and Windows with no extra setup.

Author, GateRelay: Secure Linux TCP Relay Service (github.com/vetr0s/gaterelay)

January 2026 — February 2026

- Built a production-grade TCP relay in Go forwarding public ingress traffic to a controlled backend, with connection limits, timeouts, graceful shutdown, and structured JSON logging.
- Hardened it with defense-in-depth: a non-root systemd user, sandboxing (capability dropping, syscall filtering, `ProtectSystem`), `ufw` firewall rules, and SSH key-only access, with a documented threat model.

Collaborator, Urban Crime Data Analysis (github.com/vetr0s/tucson-crime-analysis)

March 2025 — May 2025

- Ran a multivariate analysis of Tucson Police crime data against neighborhood income and streetlight locations to test how socioeconomic and environmental factors relate to crime.
- Built preprocessing, geospatial, and modeling pipelines in Python (pandas, geopandas, scikit-learn, statsmodels) using Ridge Regression, Random Forest, Logistic Regression, and OLS to predict high-crime wards.

EXTRACURRICULAR ACTIVITIES

FIRST Robotics Competition team 4450 (Safety Captain)

September 2021 — June 2023

- Led safety operations for a competitive engineering team, enforcing shop, electrical, and tooling safety standards under strict build and competition deadlines.
- Co-planned an international outreach program, coordinating risk mitigation for a \$60,000 fundraiser that sent 17 students and metal-based FTC robotics kits to Cairo, Egypt for hands-on robotics education.

Google Developer Student Club

August 2023 — May 2026

- Participated in peer-led workshops and collaborative projects on modern software development and Google Cloud technologies.

HackAZ 2026

April 2026 — April 2026

- Built and deployed GridWise Energy with 3 teammates in a 24-hour hackathon, a live web app analyzing energy usage from local power-grid data, on a React/TypeScript, Vercel, and Neon Postgres stack.
- Shipped auth, time-series usage tracking, AI chatbot insights, an ML forecast model, and an opt-in global leaderboard.